

Understanding Analysis, 2nd Edition, Stephen Abbott

Chapter 4 - Section 4.4 - Continuous Functions on Compact satisfies

1. (a) Show that $f(x) = x^3$ is continuous on all of $\mathbf{R}$.
- (b) Argue, using Theorem 4.4.5, that f is not uniformly continuous on $\mathbf{R}$.
- (c) Show that f is uniformly continuous on any bounded subset of $\mathbf{R}$.

Solution:

- (a) $f(x) = x$ is continuous on all of $\mathbf{R}$. Product of continuous functions is continuous, so $f(x) = x^3$ is continuous as well.
- (b) As per the Theorem, we need a sequence $(x_n), (y_n)$ such that $|x_n - y_n| \rightarrow 0$ but $|f(x_n) - f(y_n)| \geq \epsilon$ for a choice of ϵ . We'll choose $\epsilon = 1$. Intuitively, as x and y gets closer we place them farther and farther out where x^3 becomes extremely steep.

We'll construct a sequence such that $|x_n - y_n| = \frac{1}{n}$ for all n . We have $|f(x) - f(y)| = |x^3 - y^3| = |x - y||x^2 + xy + y^2|$. Since $|x - y| = \frac{1}{n}$, to make the expression ≥ 1 we need $|x^2 + xy + y^2| \geq n$. Simply choose $x, y \geq n$ for each n , then $|x^2 + xy + y^2| > n^2 > n$, as required. One possible sequence is $(x_n) = n$ and $(y_n) = n + \frac{1}{n}$.

- (c) Since the subset is bounded, there exists an $M > 0$ such $|x| \leq M$ for all x . The idea is that the delta for M works for all other $|x| < M$ as well, because x^3 gets steeper farther out from zero and requires smaller and smaller δ .

More formally, do the usual $\epsilon - \delta$ proof to get $|x - y||x^2 + xy + y^2|$. We have $|x|, |y| \leq M$ so $|x - y||x^2 + xy + y^2| \leq 3M^2|x - y|$. Pick $\delta = \frac{\epsilon}{3M^2}$. For an $\epsilon > 0$, this single δ works for all x .